Mini-App: 2D Viscous Burgers' Equation

CS555: Numerical Methods for PDEs — Spring 2026

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Problem Statement

Goal: Explore the interplay of advection and diffusion in numerical PDE solvers.

The **2D coupled viscous Burgers' equations**:

$$u_t + u\,u_x + v\,u_y = \nu\left(u_{xx} + u_{yy}\right)$$
$$v_t + u\,v_x + v\,v_y = \nu\left(v_{xx} + v_{yy}\right)$$

- Nonlinear advection (hyperbolic) + linear diffusion (parabolic)
- **Cole–Hopf transformation** provides an exact solution for verification
- Domain: $[0,1]^2$, Dirichlet BCs from the exact solution
- Reynolds number $\mathrm{Re} = UL/\nu$ controls solution sharpness

Mini-app highlights: Convergence rates, stability limits, cost trade-offs, and viscosity effects.

Implementation Overview

Spatial Discretization

- Central differences ($O(h^2)$)
- First-order upwind ($O(h)$)
- Laplacian via Kronecker products

Temporal Discretization

- Forward Euler — $O(\Delta t)$
- RK4 — $O(\Delta t^4)$
- Backward Euler (semi-implicit) — $O(\Delta t)$
- Crank–Nicolson with predictor-corrector — $O(\Delta t^2)$

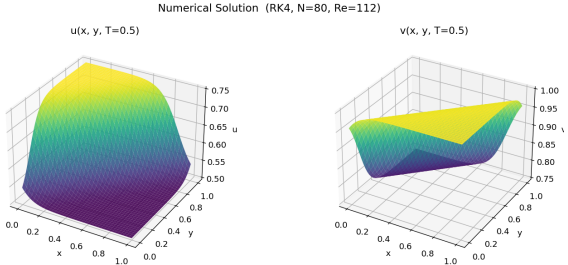
Key Design Choices

- Built from first principles in Python
- Implicit methods: sparse LU factorization (pre-factored once)
- RK4 CFL monitoring: tracks both advective and diffusive limits
- Cole–Hopf exact solution for rigorous error measurement

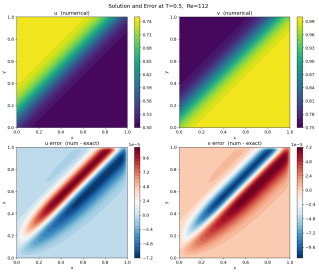
Verification Strategy

- Discrete L^2 and L^∞ norms
- Spatial & temporal convergence studies
- Stability boundary probes

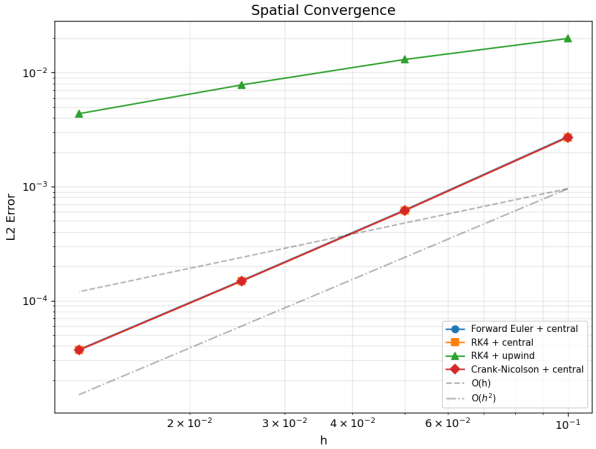
Reference Solution ($\nu = 0.01$, $\mathrm{Re} \approx 100$, $T = 0.5$)



- 81×81 grid, RK4, $\Delta t = 10^{-4}$
- Diagonal front from Cole–Hopf solution
- L^2 error $\sim 10^{-5}$
- Solution verified against exact Cole–Hopf



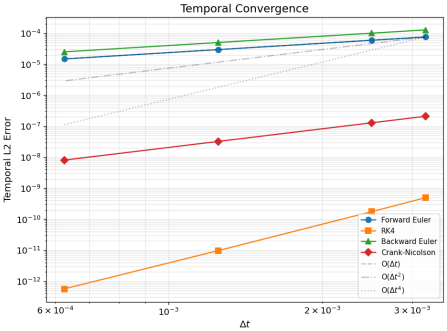
Spatial Convergence

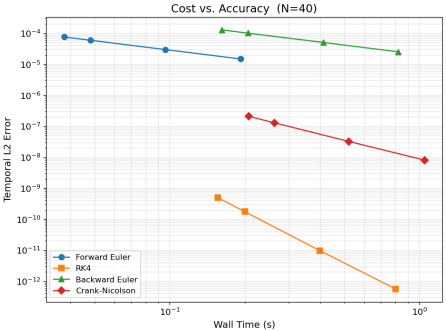


Key observations:

- Central differences $\rightarrow O(h^2)$ for all time integrators
- Upwind $\rightarrow O(h)$ due to first-order spatial truncation
- Forward Euler, RK4, CN all collapse onto the same spatial rate when Δt is small enough
- Upwind adds numerical diffusion that can mask the true solution at coarse grids

Temporal Convergence & Cost vs. Accuracy

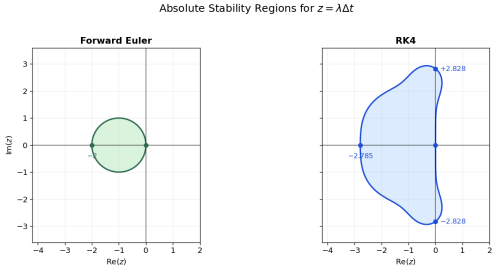


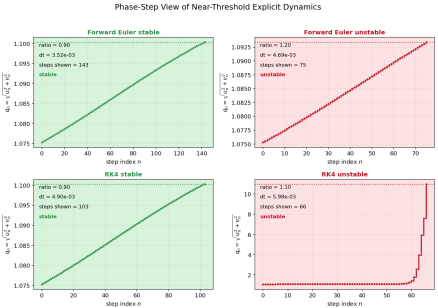


- FE, BE: $O(\Delta t)$ CN: $O(\Delta t^2)$
- RK4: $O(\Delta t^4)$

- RK4 reaches low error fastest
- Implicit methods: large Δt but expensive per step
- CN: good balance at moderate accuracy

Stability Analysis

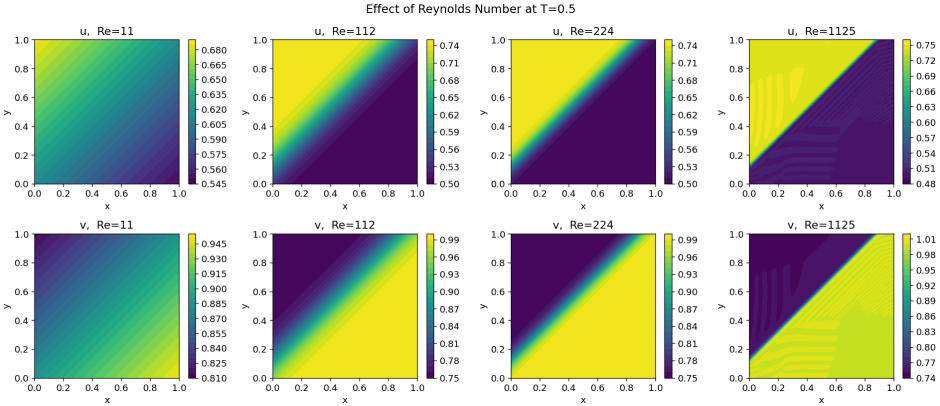




Stability regions: FE (left), RK4 (right)

- FE: $\Delta t \leq h^2/(4\nu)$ — very restrictive for fine grids
- RK4: $\approx 1.39\times$ larger stable region along the real axis
- Empirical probes confirm theoretical thresholds

Effect of Viscosity (Reynolds Number Sweep)



- As $\nu \to 0$ ($\mathrm{Re} \to \infty$), sharp gradients develop $\to$ finer grids or upwinding needed
- At $\nu = 0.001$ ($\mathrm{Re} \sim 1000$), the front resembles a shock

Reflections & Future Directions

What we learned:

1. Central vs. upwind: $O(h^2)$ accuracy vs. numerical diffusion trade-off
2. Semi-implicit splitting degrades temporal order unless carefully designed (predictor-corrector CN recovers $O(\Delta t^2)$)
3. RK4 is the most cost-effective explicit method for this problem
4. High Re exposes the need for shock-capturing or higher-order schemes

Future directions:

- WENO / flux-limiter schemes for the advection term
- Adaptive mesh refinement near sharp fronts
- Extension to 3D or more complex nonlinear PDEs (e.g., Navier–Stokes)
- Multigrid or Krylov solvers for the implicit systems at scale

Thank You

Code and notebook available in the project repository.

Questions?